DiscoveryRecommendationSkill: Individual Report Outline

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Social Network Analysis, Spring 2026
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Abstract

This outline is structured for the required three-page NeurIPS-format individual Skill report. It focuses on `DiscoveryRecommendationSkill`, the recommendation-side Skill that transforms user interest signals into arXiv query plans, candidate papers, ranked recommendations, feedback-adjusted rankings, and follow-up results. The final report should explicitly cover functionality, references, results, analysis, and visualization.

1 Functionality

Skill purpose. Define `DiscoveryRecommendationSkill` as the Skill responsible for finding and ranking papers before synthesis. It answers: which papers should be recommended, why are they ranked highly, and how should the recommendation adapt after seed papers or user feedback?

Inputs. List the exact input types:

- `RetrievalQuery`: topic, category, date range, result budget, search mode, and query-planner mode.
- Seed papers: arXiv IDs, arXiv URLs, or title text used to build a `SeedPreference`.
- Profile state: saved seed preferences and historical like/dislike feedback.
- Follow-up query: local-first filters over stored papers, with optional arXiv fetch when local results are empty.

Outputs. Describe the structured outputs: `QueryPlan`, cached `PaperMetadata`, Top-K `Recommendation` rows with rank, score, rationale, evidence label, score breakdown, feedback rank deltas, and `SkillResult` statuses for success, empty, fallback, and error cases.

2 Method and implementation

Query planning and retrieval. Explain that the Skill uses a query planner to expand user input into controlled arXiv query variants. In deterministic or fallback mode, the planner preserves required terms; in live mode, invalid LLM query plans fall back to deterministic planning. Retrieval uses the arXiv Atom API, normalizes metadata, and stores papers in SQLite for cache reuse.

Ranking. Describe two ranking paths:

- Deterministic ranking combines topic/query-plan matches, phrases, category/date context, seed preference, retrieval source metadata, and feedback events.

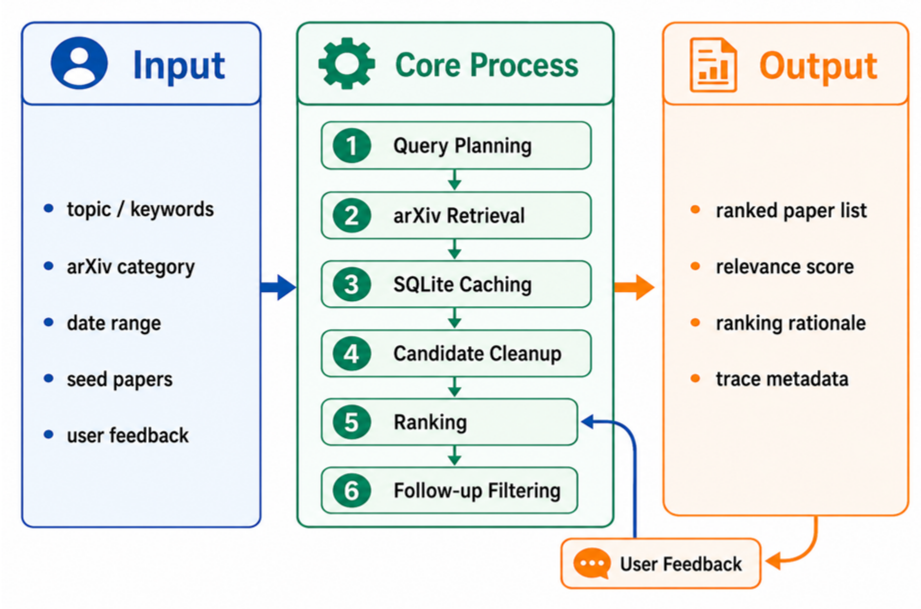

Figure 1: Planned individual-report visualization: DiscoveryRecommendationSkill workflow from user interest signals to query planning, retrieval, ranking, feedback refinement, and follow-up filtering.

- Semantic seed ranking checks embedding readiness, computes seed-candidate similarity, and fails closed when real embedding credentials or seed quality are insufficient.

State that each recommendation includes a rationale so the downstream briefing can expose why a paper was selected.

Feedback and follow-up. Explain that feedback is paper-level like/dislike. The Skill records feedback events, applies latest-wins conflict handling, and returns rank/score movement after refinement. Follow-up queries search local SQLite papers first and call arXiv only when no stored papers match and fetching is allowed.

3 References and related work

Use this section to ground the Skill in information retrieval and recommender-system methods:

- arXiv API retrieval as the source of scholarly metadata (arXiv, 2026).
- Classical information retrieval and ranked retrieval metrics such as precision, recall, and MRR (Manning et al., 2008).
- Hybrid content-based recommendation for combining topic, category, seed, and feedback signals (Burke, 2002).
- Sentence embedding similarity for semantic seed recommendation (Reimers and Gurevych, 2019).

4 Results

Report deterministic offline results from the search-quality fixture and the Unit 8 evaluation helpers. Make clear that the goal is repeatable Skill validation, not a large-scale benchmark claim.

Qualitative demo results. Use the presentation example: with topic "world model" and seed papers about "world action model," the recommendations concentrate on robotics-related world-model papers. In the profile-only demo, historical seed papers and feedback drive recommendations toward robotics and deep learning without a fresh topic.

Table 1: Planned DiscoveryRecommendationSkill results table.

Evaluation	Metric	Fixture-backed result
Broad planned search	relevant candidate coverage	1.0
Top-K ranking	precision@3 / recall@3 / MRR	1.0 / 1.0 / 1.0
Explanation support	rationale coverage	1.0
Semantic seed ablation	lexical baseline precision@K	0.0
Semantic seed ablation	semantic precision@K	1.0
Feedback refinement	reported output	rank and score movement

5 Analysis

What works and why. Argue that broad query planning improves candidate recall over strict phrase retrieval; seed papers and semantic similarity make recommendations more personalized; rationales and score breakdowns make ranking inspectable; feedback refinement gives users visible control; and local-first follow-up makes interaction faster and more contextual.

Ablations and failure cases. Include the following analyses in the final report:

- Strict phrase retrieval versus broad planned retrieval.
- Lexical-only baseline versus semantic seed ranking.
- Before/after feedback movement for liked and disliked papers.
- Local follow-up hit versus empty local store with optional fetch.

Discuss failure behavior: invalid query plans fall back, empty retrieval returns structured empty results, semantic mode fails closed, and provider payloads or raw vectors are not exposed in traces.

6 Page allocation for the final three-page report

Table 2: Suggested three-page allocation. References do not count toward the page limit.

Page target	Content
Page 1	Functionality, inputs/outputs, Figure 1
Page 2	Method, implementation details, related work
Page 3	Results, ablations, limitations, StudyClawHub metadata

Checklist before finalizing

- Replace author placeholders and add the published StudyClawHub Skill URL.
- Keep the final individual report at three pages before references.
- Include at least one visualization; Figure 1 satisfies the requirement.
- Keep recommendation evidence separate from abstract/full-text evidence.

References

arXiv. arXiv API User Manual. `https://info.arxiv.org/help/api/index.html`.

Robin Burke. Hybrid recommender systems: Survey and experiments. *User Modeling and User-Adapted Interaction*, 12:331–370, 2002.

Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schutze. *Introduction to Information Retrieval*. Cambridge University Press, 2008.

Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. *EMNLP-IJCNLP*, 2019.